

Infosys SP/DSE Coding Preparation List

Priority-wise Question Bank (Compiled from All Sources)

Test Pattern: 3 Questions / 3 Hours | Easy → Medium → Hard **Strategy:** Easy (20-30 min) → Medium (60-75 min) → Hard (remaining time)

P0: MUST DO (Highest Frequency / Actual PYQs / Easy-Medium Slot)

These appear most often in Infosys SP/DSE assessments. Master these first.

Arrays & Strings (Easy Slot)

#	Problem	Topic	Difficulty	Source
1	Two Sum (Pair with given Sum)	HashMap / Two Pointers	Easy	GFG SDE Sheet + Dee_codes
2	Best Time to Buy and Sell Stock	Array Traversal	Easy	GFG SDE Sheet + Dee_codes
3	Majority Element	HashMap / Boyer-Moore	Easy	Dee_codes priority list
4	Valid Anagram	HashMap / Sorting	Easy	Dee_codes priority list
5	Product of Array Except Self	Prefix Sum	Easy-Medium	Dee_codes priority list
6	Remove Duplicates from Sorted Array	Two Pointers	Easy	GFG SDE Sheet (listed twice)
7	Reverse a String	Two Pointers	Easy	GFG SDE Sheet
8	Check if Array is Sorted	Linear Scan	Easy	GFG SDE Sheet
9	Maximum Subarray Sum (Kadane's Algorithm)	DP / Greedy	Easy-Medium	GFG SDE Sheet + Let's Code (Very High Freq)
10	First Non-Repeating Character	HashMap / String	Easy	Let's Code (Very High Freq)
11	Element Frequency Analysis (Repeated & Non-repeated count)	HashMap	Easy	Let's Code (Very High Freq)
12	Maximum Vacation Days (Max consecutive days between obligations)	Array / Sorting	Easy-Medium	Let's Code (High Freq)
13	Binary Search	Divide & Conquer	Easy	GFG SDE Sheet
14	First Occurrence in Sorted Array	Binary Search	Easy-Medium	Let's Code (High Freq)
15	Count Frequencies in an Array	HashMap	Easy-Medium	GFG SDE Sheet

Table 1 – continued

#	Problem	Topic	Difficulty	Source
16	Subarray with Given Sum	Sliding Window / Prefix Sum	Easy-Medium	GFG SDE Sheet
17	Rotate Array	Array Manipulation	Easy-Medium	GFG SDE Sheet
18	Next Permutation	Array	Medium	GFG SDE Sheet
19	Merge Two Sorted Arrays (Without Extra Space)	Two Pointers	Medium	GFG SDE Sheet
20	Find element that appears once in sorted array	Binary Search	Medium	GFG SDE Sheet

Greedy (Medium Slot)

#	Problem	Topic	Difficulty	Source
21	Activity Selection	Greedy / Sorting	Medium	Dee_codes
22	Assign Cookies	Greedy / Two Pointers	Medium	Dee_codes
23	Fractional Knapsack	Greedy / Sorting	Medium	Dee_codes
24	Minimum Platforms Required	Greedy / Sorting	Medium	Dee_codes + GFG + GitHub (Infosys-SP)
25	Minimum Coins Required (Coin Change - min coins)	Greedy / DP	Medium	GitHub (Infosys-SP)
26	Job Sequencing Problem	Greedy	Medium	GFG SDE Sheet
27	Max Length Chain	Greedy	Medium	GFG SDE Sheet
28	Largest Number formed from an Array	Greedy / Custom Sort	Medium	GFG SDE Sheet + GitHub (Infosys-SP)
29	Array Minimization Problem (Subarray subtract vs direct zero)	Greedy / Prefix Sum	Medium	Let' s Code (High Freq PYQ)
30	Wine Bottle Transport (Min work to fulfill demands)	Greedy / Prefix Sum	Medium	GitHub (Infosys-SP)
31	Unique Birthday Gift	Greedy	Medium	GitHub folder
32	Largest Number (Arrange strings for max concatenated number)	Greedy / Custom Sort	Medium	GitHub (Infosys-SP)

Table 2 – continued

#	Problem	Topic	Difficulty	Source
33	Minimum Imbalance of Pedestals	Prefix Sum + Sorting	Easy	Dee_codes (actual PYQ style)
34	Neutralize Zone (Sort desc, pick top K)	Sorting	Easy-Medium	Dee_codes (actual PYQ style)
35	Food Stamps (Max taste points with decay)	Greedy / Priority Queue	Easy-Medium	External Doc (Infosys Limited)
36	Minimize the Heights II	Greedy	Medium	GFG SDE Sheet
37	Minimum number of jumps	Greedy / DP	Medium	GFG SDE Sheet
38	Generate Parentheses	Backtracking / Recursion	Medium	GFG SDE Sheet
39	House Robber	DP / Greedy	Medium	GFG SDE Sheet + Dee_codes
40	Allocate minimum number of pages	Binary Search + Greedy	Medium	GFG SDE Sheet

Dynamic Programming (Hard Slot)

#	Problem	Topic	Difficulty	Source
41	0/1 Knapsack Problem	DP	Medium-Hard	Dee_codes + GFG + Let' s Code + GitHub
42	Climbing Stairs	DP	Easy	Dee_codes
43	Coin Change (ways/min coins)	DP	Medium-Hard	Dee_codes + GitHub
44	Subset Sum	DP	Medium	GFG SDE Sheet + GitHub (Infosys-SP)
45	Longest Common Subsequence (LCS)	DP	Hard	GFG SDE Sheet
46	Longest Increasing Subsequence (LIS)	DP / Binary Search	Medium-Hard	GFG SDE Sheet + GitHub folder
47	Edit Distance	DP	Hard	GFG SDE Sheet
48	Longest Palindromic Substring	DP / Expand Around Center	Medium-Hard	GFG SDE Sheet
49	Gift Box Packing Problem (Max distinct types in K consecutive boxes)	DP + Sliding Window	Medium-Hard	Let' s Code (High Freq PYQ)
50	MSS with Swaps (Max subarray sum after exactly K swaps)	DP / Greedy	Medium-Hard	External Doc (Infosys Limited)

P1: HIGH PRIORITY (Medium-Hard / Frequent Patterns)

These solidify your medium and hard slot preparation.

#	Problem	Topic	Difficulty	Source
51	Container With Most Water	Two Pointers	Medium	Dee_codes
52	Longest Substring Without Repeating Characters	Sliding Window	Medium	Dee_codes
53	Maximum Sum Subarray of Size K	Sliding Window	Easy-Medium	Dee_codes
54	Same Digit Base Conversion (Find base K where M has identical digits)	Number Theory	Medium	Let' s Code (Medium Freq PYQ)
55	Reverse Linked List	Linked List	Easy-Medium	Let' s Code (Very High Freq)
56	Add Two Numbers (Linked Lists)	Linked List + Math	Medium	Let' s Code (High Freq)
57	Longest Common Substring	DP	Hard	GFG SDE Sheet
58	Matrix Chain Multiplication	DP	Hard	GFG SDE Sheet
59	Partition Equal Subset Sum	DP	Hard	GFG SDE Sheet
60	Wildcard Pattern Matching	DP	Hard	GFG SDE Sheet
61	Form a Palindrome (Min insertions)	DP	Hard	GFG SDE Sheet
62	Minimum Cost Path	DP / Greedy	Medium	GFG SDE Sheet
63	Number of subarrays having sum exactly equal to K	Prefix Sum + HashMap	Medium	GFG SDE Sheet
64	Find smallest positive integer not representable as subset sum	Greedy	Medium	GFG SDE Sheet
65	Terrain Transformation (Make strictly decreasing using powers of 2)	Binary Search + Greedy	Hard	Let' s Code (Medium Freq PYQ)

Table 4 – continued

#	Problem	Topic	Difficulty	Source
66	Lock & Parity (Min cost assignment with even odd parity)	Greedy / Graph Matching	Hard	External Doc (Infosys Limited)
67	Bitwise AND of Range [L,R]	Bit Manipulation	Medium-Hard	Let' s Code (Low Freq)
68	Array and Bit Manipulation (Smallest power of 2 $\geq$ x)	Bit Manipulation	Medium	Let' s Code (Medium Freq)
69	Bitwise Subsequence	Greedy / Bitwise	Medium	GitHub folder
70	Max XOR Subset	Greedy / Bitwise	Medium	GitHub folder
71	Conditional LIS (LIS with bitwise condition between adjacent)	DP + Bitwise	Hard	Let' s Code (Low Freq)
72	Detect Cycle in a Linked List	Floyd' s Cycle	Medium	GFG SDE Sheet
73	Max Rectangle in Binary Matrix	DP / Histogram	Hard	GFG SDE Sheet
74	Count Smaller elements	Merge Sort / BIT	Hard	GFG SDE Sheet
75	Minimum number of deletions to make palindrome	DP (LCS variant)	Hard	GFG SDE Sheet

P2: IMPORTANT (Hard / Edge Cases / Interview Depth)

For Hard slot partial solutions and technical interview follow-ups.

#	Problem	Topic	Difficulty	Source
76	Layer-Split Path Maximization with Penalties	Graph + DP	Complex	External Doc (Infosys Limited)
77	Find all distinct subset (or subsequence) sums	DP / Backtracking	Hard	GFG SDE Sheet
78	Probability of Knight	DP / BFS	Hard	GFG SDE Sheet
79	Tower of Hanoi	Recursion	Medium	Let' s Code (Low Freq)
80	Factorial Calculation (Large number handling)	Basic / Math	Easy	Let' s Code (Medium Freq)

Table 5 – continued

#	Problem	Topic	Difficulty	Source
81	Octal to Binary Conversion	Number System	Easy	Let' s Code (Medium Freq)
82	Goal Tracker (Average steps per day)	Basic Math	Easy	Let' s Code (Medium Freq)
83	Swap Two Arrays	Array Manipulation	Easy	Let' s Code (Low Freq)
84	Print 1 to N Without Using Loops	Recursion	Easy	GFG SDE Sheet
85	Sum of all prime numbers between 1 and N	Math / Sieve	Easy	GFG SDE Sheet
86	Armstrong Numbers	Math	Easy	GFG SDE Sheet
87	Perfect Number	Math	Easy	GFG SDE Sheet
88	Sum Palindrome	Math / String	Easy	GFG SDE Sheet
89	Form largest number from digits	Sorting	Easy	GFG SDE Sheet
90	Number Of Open Doors	Math	Easy	GFG SDE Sheet
91	Overlapping Rectangles	Math / Geometry	Easy	GFG SDE Sheet
92	Check if four points form a square	Math / Geometry	Easy	GFG SDE Sheet
93	Maximum Possible Sum of Products	Greedy / Sorting	Easy	GFG SDE Sheet
94	Common Divisors	Math	Easy	GFG SDE Sheet
95	Check if number is power of another number	Math	Easy	GFG SDE Sheet
96	Party of Couples (Find single person)	XOR / HashMap	Easy	GFG SDE Sheet
97	Tidy Number	String / Math	Easy	GFG SDE Sheet
98	Print the Left Element	Simulation	Easy	GFG SDE Sheet
99	Middle of Three	Comparison	Easy	GFG SDE Sheet
100	Binary Representation	Bitwise	Easy	GFG SDE Sheet
101	Greatest of Three Numbers	Basic	Easy	GFG SDE Sheet
102	Celsius to Fahrenheit Conversion	Formula	Easy	GFG SDE Sheet
103	Convert from Any Base to Decimal	Number System	Easy	GFG SDE Sheet

Table 5 – continued

#	Problem	Topic	Difficulty	Source
104	Check if two matrices are identical	Matrix	Easy	GFG SDE Sheet
105	Addition of Two Matrices	Matrix	Easy	GFG SDE Sheet
106	Area of Rectangle, Triangle, Circle	Formula	Easy	GFG SDE Sheet
107	Sort String of Characters	Sorting	Easy	GFG SDE Sheet
108	Sentence Palindrome	String	Easy	GFG SDE Sheet
109	Remove common characters and concatenate	String / HashSet	Easy	GFG SDE Sheet
110	Check if string is rotated by two places	String	Easy	GFG SDE Sheet
111	Check if Date Is Valid	String / Logic	Easy	GFG SDE Sheet
112	The Lazy Caterer' s Problem	Math	Easy	GFG SDE Sheet
113	Program to Print Sum Triangle for a Given Array	Recursion	Easy	GFG SDE Sheet
114	Count Digits in a Number	Basic	Easy	GFG SDE Sheet
115	Print First N Fibonacci Numbers	Basic / DP	Easy	GFG SDE Sheet

P3: GOOD TO HAVE (Complex / HackWithInfy R2 / Low Probability)

Only if you have time after P0-P2. These are for HackWithInfy Round 2 or Power Programmer track.

#	Problem	Topic	Difficulty	Source
116	Graph BFS/DFS Basics	Graph Traversal	Medium	Dee_codes
117	Dijkstra' s Shortest Path	Graph + Heap	Hard	HackWithInfy R2
118	Union-Find / DSU	Graph	Hard	HackWithInfy R2
119	Segment Tree / Fenwick Tree (BIT)	Range Queries	Hard	HackWithInfy R2

Table 6 – continued

#	Problem	Topic	Difficulty	Source
120	Trie (Prefix Tree)	Strings	Hard	HackWithInfy R2
121	DP combined with Graphs	Advanced DP	Complex	HackWithInfy R2

Data Structures You Must Be Comfortable With

Based on your HackWithInfy experience (ArrayList/LinkedList freeze), here are the DS ranked by Infosys frequency:

Rank	Data Structure	Why It Matters	Infosys Frequency
1	Array / String	Easy slot questions, foundation of everything	Very High
2	HashMap / HashSet	Two Sum, Anagram, Frequency, Distinct elements	Very High
3	ArrayList (Java)	Dynamic arrays, list of lists, 2D structures	Very High
4	LinkedList	Reverse, Add Two Numbers, cycle detection	High
5	Stack / Queue	Parentheses, BFS, basic implementations	Medium
6	Tree / BST	BFS/DFS basics, traversal (Dee_codes mentions)	Medium
7	PriorityQueue / Heap	Food Stamps, merge problems, greedy	Medium
8	LinkedHashMap	First non-repeating character (insertion order)	Low-Medium
9	Graph (Adjacency List)	BFS/DFS, Dijkstra (R2 only)	Low
10	Trie / Segment Tree	Only for HackWithInfy R2 / Power Programmer	Very Low

Java Collections Traps to Fix (Your Specific Gap)

1. **ArrayList vs LinkedList API:** `addFirst()/addLast()` vs `get(i)` — know when $O(1)$ vs $O(n)$
2. **ListIterator:** Use for removal during iteration, not for-each
3. **List<List>:** Must instantiate inner lists separately: `new ArrayList<>()` for each row
4. **Queue/Deque interface:** `ArrayDeque` is preferred over `LinkedList` for stack/queue ops
5. **PriorityQueue:** Default is min-heap; for max-heap use `Collections.reverseOrder()`
6. **HashMap iteration:** `entrySet()` for key-value pairs, not separate `keySet()+get` loops

Recommended Prep Order (Time-boxed)

Week 1: P0 Easy (Arrays/Strings/Hashing)

- Days 1-2: Two Sum, Best Time to Buy/Sell, Majority Element, Valid Anagram, Product Except Self
- Days 3-4: Remove Duplicates, Reverse String, First Non-Repeating, Element Frequency, Max Vacation Days
- Days 5-6: Kadane's, Binary Search variants, First Occurrence, Subarray Sum
- Day 7: Mock Easy test (3 questions, 30 min each)

Week 2: P0 Greedy (Medium Slot)

- Days 1-2: Activity Selection, Assign Cookies, Fractional Knapsack, Minimum Platforms
- Days 3-4: Job Sequencing, Max Length Chain, Largest Number, Array Minimization
- Days 5-6: Wine Selling, Food Stamps, House Robber, Minimize Heights
- Day 7: Mock Medium test (2 questions, 75 min each)

Week 3: P0-P1 DP (Hard Slot)

- Days 1-2: Climbing Stairs, Coin Change, 0/1 Knapsack, Subset Sum
- Days 3-4: LCS, LIS, Edit Distance, Longest Palindromic Substring
- Days 5-6: Gift Box Packing, MSS with Swaps, Lock & Parity, Partition Equal Subset
- Day 7: Mock Hard test (1 question, 90 min) + Full mock (3 questions, 3 hours)

Week 4: P1-P2 + Java Collections Fix

- Days 1-2: Linked List (Reverse, Add Two Numbers, Cycle Detection)
- Days 3-4: Bitwise (AND of Range, Power of 2, Max XOR, Bitwise Subsequence)
- Days 5-6: Java Collections drill — 8-problem set forcing ArrayList/LinkedList/HashMap/Queue translation
- Day 7: Final full mock under timed conditions

Key Reminders

- **Realistic target:** Easy fully solved + Medium fully solved + Hard partial = Competitive
- **Don't spend >30 min on Easy** — if stuck, brute force first, optimize later
- **Edge cases:** Empty arrays, single element, all negatives, duplicates, overflow (use long)
- **Partial Hard solution:** Write brute force first, then optimize — partial marks > no submission